Bihar Engineering University, Patna

B.Tech 3rd Semester Examination, 2024

Course: B.Tech | *Time: 03 Hours*
Code: 100306 | *Subject: Electrical Circuit Analysis* | *Full Marks: 70*

Instructions: -

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct answer from the following (Any seven question only): **[2 x 7 = 14]**

(a) Thevenin's and Norton's equivalents are:
- (i) Different representations with different values
- (ii) Identical in all circuits
- (iii) Equivalent with same behavior at load terminals
- (iv) Applicable only for AC circuits

(b) Mesh analysis is preferred when:
- (i) The number of meshes is more than nodes
- (ii) The number of meshes is less than nodes
- (iii) Dependent sources are present
- (iv) Circuits are complex

(c) The steady-state response in a circuit is observed:
- (i) During initial $t = 0$
- (ii) When $t \to \infty$
- (iii) When energy is maximum
- (iv) Just after switching

(d) In a critically damped RLC circuit, the roots of characteristic equation are:
- (i) Complex conjugates
- (ii) Real and distinct
- (iii) Real and equal
- (iv) Purely imaginary

(e) The admittance of an inductor is:
- (i) $j\omega L$
- (ii) $1/(j\omega L)$
- (iii) $-j\omega L$
- (iv) $j\omega C$

(f) The average power in a purely capacitive circuit is:
- (i) Zero
- (ii) Maximum
- (iii) Depends on frequency
- (iv) Infinite

(g) Poles are values of 's' where:
- (i) Input is zero
- (ii) Output is zero
- (iii) Denominator of transfer function is zero
- (iv) Numerator is zero

(h) Convolution in time domain becomes.........in Laplace domain:
- (i) Division
- (ii) Addition
- (iii) Multiplication
- (iv) Integration

(i) A network is reciprocal if it contains only:
- (i) Resistors
- (ii) Resistors and capacitors
- (iii) Passive linear elements
- (iv) Active devices

(j) The h-parameters are mainly used for:
- (i) Filter design
- (ii) Inductor modeling
- (iii) Transistor modeling
- (iv) Transformer design

Q.2 (a) State Maximum Power Transfer Theorem. Prove that efficiency of the circuit under maximum power transfer condition is 50%. [7]

(b) Find the voltage across 3Ω resistor using any theorem for the circuit shown in Figure-1 [7]

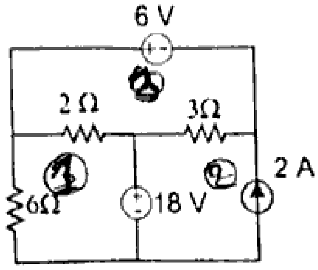

Figure 1

Q.3 (a) Find V_A using the principle of superposition in network as shown in Figure 2. [7]

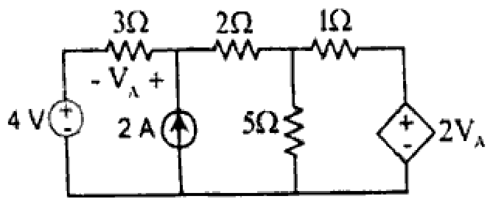

Figure 2

(b) State Reciprocity Theorem. Verify the Reciprocity Theorem for the circuit shown in Figure 3. [7]

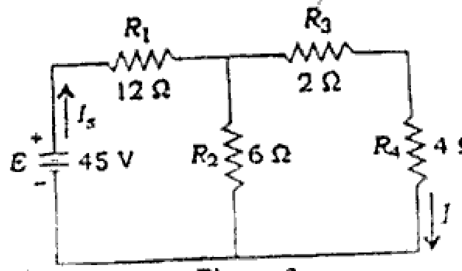

Figure 3

Q.4 (a) Explain the behaviour of RL and RC elements for transient condition. Mention their representation at the time of switching. [7]

(b) Draw the circuit of parallel RLC circuit and derive the formulae for resonant frequency, half-power frequencies, bandwidth and quality factor. [7]

Q.5 (a) Determine Laplace transform of a following (i) sin2t (ii) cos2t [7]

(b) Synthesis the waveform shown in Figure 4 below and Determine Laplace transform of periodic waveform. [7]

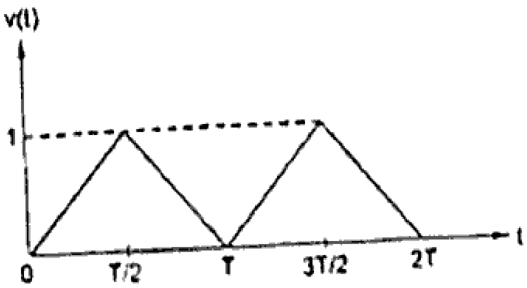

Figure 4

Q.6 (a) To find Thevenin equivalent between terminals 'a' and 'b' as shown in Figure 5. [7]

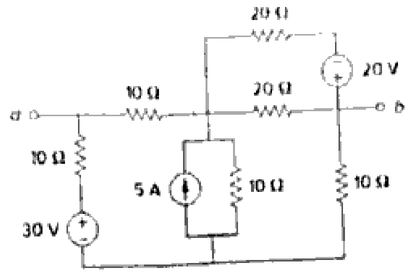

Figure 5

(b) A switch is moved from position 1 to 2 at t=0 as shown in Figure 6. Find $V_R(t)$ and $V_C(t)$ for t>0. [7]

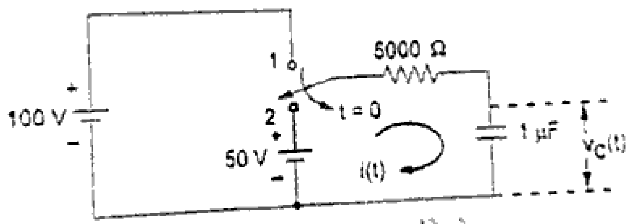

Figure 6

Q.7 (a) Explain about the transmission parameters and derive the condition for symmetry and reciprocity. [7]

(b) Determine z-parameter of network shown below in Figure 7 & give its equivalent circuit. [7]

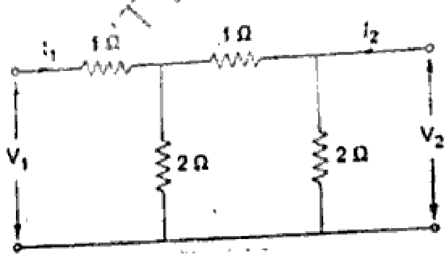

Figure 7

Q.8 (a) Define two-port network and write assumptions to be made to find network and also obtain Y parameter. [7]

(b) Two coils connected in series have an equivalent inductance of 10H. When the connections of one coil are reversed, the effective inductance is 6H. If the co-efficient of coupling is 0.6, calculate the self-inductance of each coil and mutual inductance. [7]

Q.9 Write short notes on *any two* of the following: [7 x 2 = 14]

(a) Poles and Zeros
(b) Convolution integral
(c) Mutual coupled circuits
(d) Transmission and Hybrid parameters